Zero-Knowledge Proofs

CS 171: Cryptography

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Overview

1. Recall
2. Three-Coloring in Zero-Knowledge
3. Zero-Knowledge: The Simulator
4. Graph Hamiltonicity
5. If Time: Schnorr, Revisited
6. Summary

Zero-Knowledge Proofs: Recall

The prover $P(1^n, x, w)$ convinces the verifier $V(1^n, x)$ that $\exists w : \ C(x, w) = 1$.

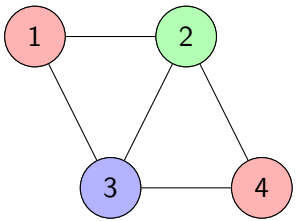
- **Completeness**: if $C(x, w) = 1$, V accepts with probability $1 - \mathrm{negl}(n)$.
- **Soundness**: if no w works, every cheating P^* convinces V with probability only $\mathrm{negl}(n)$.
- **Zero-knowledge**: a simulator $\mathcal{S}$ without w produces a view indistinguishable from that of any PPT V^* (game ZK).

Our tool: commitments (lecture 20)

Binding: $\mathrm{Com}(x; r)$ can only be opened to x. **Hiding**: it reveals nothing about x.

Graph Three-Coloring

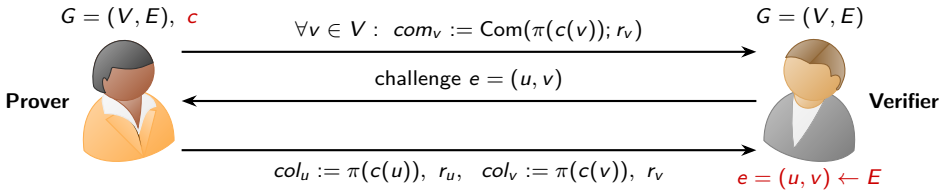
- A graph $G = (V, E)$.
- Task: find $c : V \to \{1, 2, 3\}$ with $c(u) \neq c(v)$ for every edge $(u, v) \in E$.



$c(1) = c(4) = 1$
$c(2) = 2$
$c(3) = 3$

- Deciding three-colorability is NP-complete.
- **As a statement**: $x = G$, $w = c$; $C(G, c) = 1$ iff $c(u) \neq c(v)$ on every edge.

A Zero-Knowledge Proof for Three-Coloring



Why shuffle? Without π, each run reveals $c(u), c(v)$. With a fresh π, the verifier sees only two distinct random colors.

Three-Coloring: Completeness and Soundness

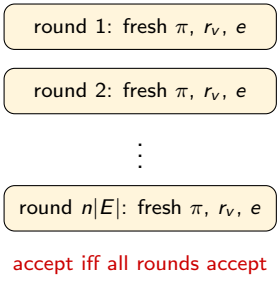
- **Completeness**: $c(u) \neq c(v)$, so $\pi(c(u)) \neq \pi(c(v))$, and the verifier accepts.
- **Soundness**: let G be **not** three-colorable. By binding, each com_v fixes a value col_v.
  - These values are not a valid three-coloring, so some edge fails the check.
  - The verifier picks that edge with probability $\geq 1/|E|$.

  So a cheating prover succeeds with probability at most $1 - \frac{1}{|E|}$.

Not negligible!

Cheating on one edge usually goes unnoticed. We need to **amplify** soundness.

Soundness Amplification

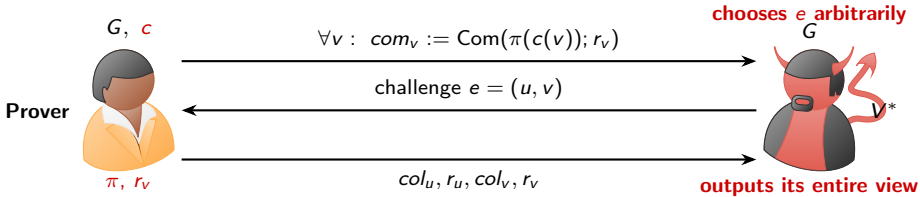


- Repeat $n|E|$ times, sequentially, with fresh randomness.
- A cheater passes each round with probability $\leq 1 - \frac{1}{|E|}$.
- It passes all $n|E|$ rounds with probability

$$\leq \left(1 - \frac{1}{|E|}\right)^{n|E|} \leq e^{-n} \quad (\text{as } 1 - z \leq e^{-z}).$$

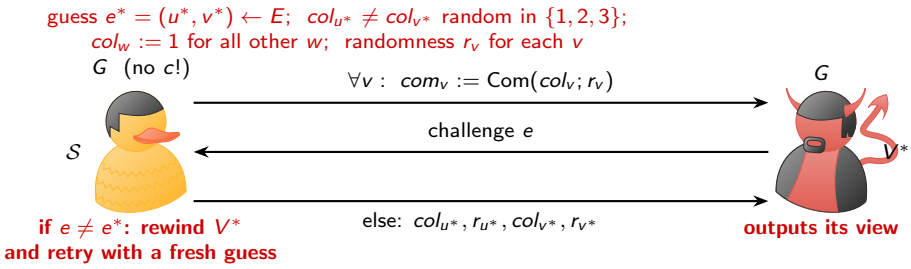
- **Completeness**: unaffected.
- **Zero-knowledge**: preserved by sequential repetition (simulate round by round).

A Malicious Verifier



- V^* may choose e based on the commitments, and output anything.
- Its **view**: input G, random coins, and all messages received.
- $\mathcal{S}$ must produce an indistinguishable view **without** c.

The Simulator $\mathcal{S}$



- By hiding, V^* cannot tell which edge was guessed: $\Pr[e = e^*] \approx 1/|E|$. If $e = e^*$, the view looks real.
- About $|E|$ tries are needed; after $n|E|$ failures, $\mathcal{S}$ outputs $\perp$ (probability $\approx e^{-n}$).

Simulation by Cropping Undesirable Parts



- Great skill?
- Took 156 attempts — the failed ones were cropped.
- Looks like the real thing — exactly what $\mathcal{S}$ does with its restarts.

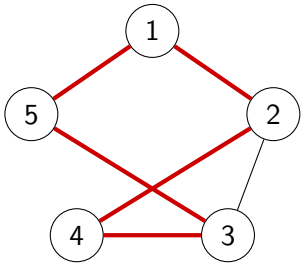
The Simulator's Output Is Indistinguishable

	V^* interacts with ...	why it looks the same
H_0	the honest prover (real view)	
H_1	... who also guesses $e^* \leftarrow E$, and **restarts** if $e \neq e^*$	identical: e^* is independent of V^*'s view (cropping)
H_2	... and commits to 1 at all vertices but u^*, v^*	**hiding**: these are never opened
H_3	the simulator $\mathcal{S}$: random $col_{u^*} \neq col_{v^*}$, no c	identical: a random π makes $\pi(c(u^*)) \neq \pi(c(v^*))$ a random pair

- $H_0 \approx H_3$: **zero-knowledge**. □
- $H_1 \to H_2$ changes one commitment at a time: a hybrid argument, as in lecture 6.

Graph Hamiltonian Cycle Problem

- $G = (V, E)$ with $V = \{1, \ldots, m\}$, as an $m \times m$ **adjacency matrix** M: $M_{u,v} = 1$ iff $\{u, v\} \in E$.
- Task: find a **Hamiltonian cycle** $\mathcal{H}$ — a cycle visiting every vertex **exactly once**.



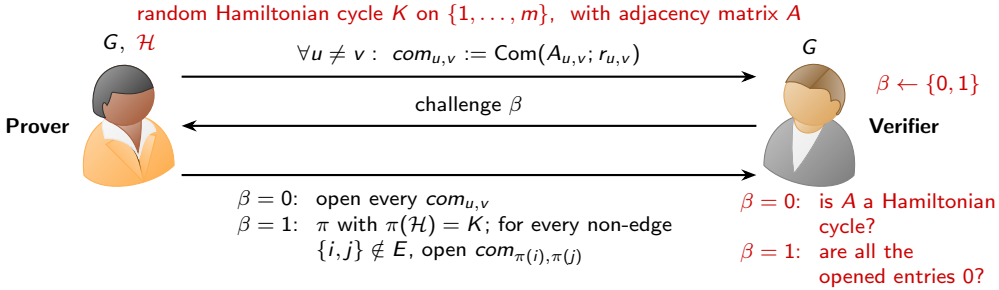
$M =$

	1	2	3	4	5
1	0	**1**	0	0	**1**
2	**1**	0	1	**1**	0
3	0	1	0	**1**	**1**
4	0	**1**	**1**	0	0
5	**1**	0	**1**	0	0

edges of $\mathcal{H}$:
$1 \to 2 \to 4 \to 3 \to 5 \to 1$

- Deciding Hamiltonicity is NP-complete.
- **As a statement**: $x = G$, $w = \mathcal{H}$; C checks that $\mathcal{H}$ is a Hamiltonian cycle of G.

A Zero-Knowledge Proof for Hamiltonicity



The first message is independent of G: a random cycle needs only m.

What the Verifier Sees

Committed

–				
	–			
		–		
			–	
				–

commitments to the matrix A of a random cycle K

$\beta = 0$**: open all**

–	0	1	1	0
0	–	0	1	1
1	0	–	0	1
1	1	0	–	0
0	1	1	0	–

$K: \ 1 \to 3 \to 5 \to 2 \to 4 \to 1$
check: a Hamiltonian cycle

$\beta = 1$**:** π**, and the non-edges**

–	**0**			**0**
0	–			
		–	**0**	
		0	–	**0**
0			**0**	–

$\pi: \ 1 \mapsto 1, \ 2 \mapsto 3, \ 3 \mapsto 2, \ 4 \mapsto 5, \ 5 \mapsto 4$
non-edges $13, 14, 25, 45 \ \mapsto \ 12, 15, 34, 45$
check: all 0

- $\beta = 0$: a random Hamiltonian cycle — unrelated to G and $\mathcal{H}$.
- $\beta = 1$: a random π and zeros at the non-edges of $\pi(G)$ — computable from G alone.
- K is fixed **before** β: passing both checks means $K \subseteq \pi(G)$, so $\pi^{-1}(K)$ is a Hamiltonian cycle of G.

Hamiltonicity: Completeness, Soundness, Zero-Knowledge

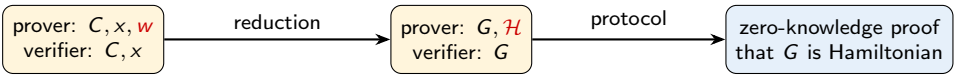
- **Completeness**: K is a Hamiltonian cycle, and $K = \pi(\mathcal{H}) \subseteq \pi(G)$, so every non-edge of $\pi(G)$ is a 0 of A.
- **Soundness**: by binding, the commitments fix A. If G is not Hamiltonian, A cannot pass both checks: a Hamiltonian cycle $K \subseteq \pi(G)$ would give the cycle $\pi^{-1}(K)$ of G. So the verifier rejects with probability $\geq \frac{1}{2}$; n repetitions give error 2^{-n}.
- **Zero-knowledge**: $\mathcal{S}$ guesses β — for 0, it commits to a random Hamiltonian cycle; for 1, it picks a random π and commits to all zeros — and restarts on a wrong guess. Hiding covers the unopened entries.

Extending to Any Computation

- Hamiltonicity is **NP-complete**: from C and x, one can build, in polynomial time, a graph G with

$$G \text{ has a Hamiltonian cycle} \iff \exists w : \; C(x, w) = 1.$$

- The reduction also maps w to a Hamiltonian cycle $\mathcal{H}$ of G, efficiently.

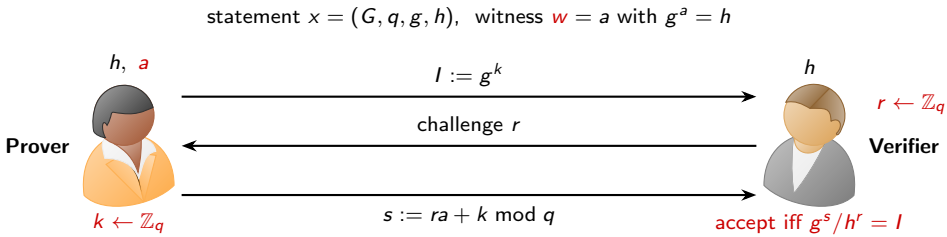


- **Every NP statement** — e.g. "this ciphertext encrypts a valid vote — has a zero-knowledge proof (given commitments, e.g. from one-way functions).

But

The proof is as long as the graph: $\mathrm{poly}(|C|)$, times n repetitions. Can a proof be much **shorter** than the computation?

Schnorr Identification Is a Proof



- Lecture 16's identification scheme, with $sk = a$. **Completeness**: $g^{ra+k}/g^{ar} = g^k$.
- But every $h \in G$ has a discrete log: "$\exists a : g^a = h$" is always **true**. So what does the verifier learn?

Proofs of Knowledge

- The verifier learns that the prover **knows** a witness — not just that one exists.
- **Proof of knowledge**: an efficient **extractor** can compute a witness from any prover that convinces the verifier, by **rewinding** it.
- Schnorr (lecture 16): rewind to just after I and ask two challenges $r_1 \neq r_2$. Two accepting answers give

$$a = \frac{s_1 - s_2}{r_1 - r_2} \bmod q.$$

- Hamiltonicity too: answers to $\beta = 0$ and $\beta = 1$ for the same K give K and π, hence the cycle $\pi^{-1}(K)$.

Is Schnorr Zero-Knowledge?

- Lecture 16's simulator: pick $r, s \leftarrow \mathbb{Z}_q$ and set $I := g^s / h^r$. Same distribution as a real transcript.
- It chooses r **before** I. That fools the **honest** verifier, whose r is random and independent of I: **honest-verifier zero-knowledge**.
- A malicious V^* may compute r from I. Guess-and-restart then succeeds with probability only $1/q$ per try.
- **Fix**: challenges $r \in \{0, 1\}$, repeated n times. A guess is right with probability $\frac{1}{2}$, so restarting works — as for Hamiltonicity.

That is why our protocols use one-bit challenges

The simulator must be able to **guess** the challenge.

Summary

- **Three-coloring**: commit to a shuffled coloring, open one random edge.
- **Soundness**: a cheater is caught with probability $\geq 1/|E|$; repeat $n|E|$ times for error $\leq e^{-n}$.
- **Simulator**: guess the edge, **restart** on a wrong guess; indistinguishable by hiding.
- **Hamiltonicity**: commit to a random cycle (independent of G); open all of it, or show it fits in $\pi(G)$.
- **NP-completeness**: every statement $\exists w : C(x, w) = 1$ has a zero-knowledge proof.
- **Schnorr**: a proof of **knowledge** of a discrete log; honest-verifier zero-knowledge.